

Set Operations in SQL

Q. Discuss the SQL Set Operations. Explain UNION, UNION ALL, INTERSECT, and EXCEPT with examples, and list the strict rules required to use them.

[DEF] Definition to Remember

SQL Set Operations allow you to combine the result sets of two or more independent SELECT queries into a single output. They are based on mathematical set theory and include UNION, UNION ALL, INTERSECT, and EXCEPT/MINUS. All require Union Compatibility between the queries.

1. Rules for Set Operations (Union Compatibility)

Both SELECT queries must satisfy:

2. Types of Set Operations

1. UNION

Result: All unique cities from both tables.

2. UNION ALL

Result: All city entries, including repeated cities.

3. INTERSECT

Result: Only cities with both a Customer AND a Supplier.

4. EXCEPT / MINUS

Result: Cities with Customers but NO Suppliers.

3. Quick Comparison

Operation	Duplicates	Returns
UNION	Removed	All from A + All from B
UNION ALL	Kept	All from A + All from B (with duplicat
INTERSECT	Removed	Common rows only
EXCEPT/MINUS	Removed	Rows in A not in B

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Set operations combine results of two SELECT queries into a single result set.
2. Union Compatibility required: same number of columns AND compatible data types.
3. UNION: combines all rows, removes duplicates.
4. UNION ALL: combines all rows, keeps duplicates (faster than UNION).
5. INTERSECT: returns only common rows present in both queries.
6. EXCEPT (MINUS in Oracle): returns rows in first query not found in the second.
7. UNION ALL is preferred for performance when duplicates are acceptable.

[RECALL] Quick Recall

Set Ops -> Union Compatibility -> UNION (no duplicates) -> UNION ALL (keeps duplicates, faster) -> INTERSECT (common only) -> EXCEPT/MINUS (A not in B)